

Problem 1. (All Russian Olympiad 2017 P8) Every cell of 100×100 table is colored black or white. Every cell on table border is black. It is known, that in every 2×2 square there are cells of two colors. Prove, that exist 2×2 square that is colored in chess order.

Problem 2. (2019 ELMO Shortlist C2) Adithya and Bill are playing a game on a connected graph with $n > 2$ vertices, two of which are labeled A and B , so that A and B are distinct and non-adjacent and known to both players. Adithya starts on vertex A and Bill starts on B . Each turn, both players move simultaneously: Bill moves to an adjacent vertex, while Adithya may either move to an adjacent vertex or stay at his current vertex. Adithya loses if he is on the same vertex as Bill, and wins if he reaches B alone. Adithya cannot see where Bill is, but Bill can see where Adithya is. Given that Adithya has a winning strategy, what is the maximum possible number of edges the graph may have? (Your answer may be in terms of n .)

Problem 3. (2011 Indonesia TST stage 2 test 3 p3) Given a board consists of $n \times n$ unit squares ($n \geq 3$). Each unit square is colored black and white, resembling a chessboard. In each step, TOMI can choose any 2×2 square and change the color of every unit square chosen with the other color (white becomes black and black becomes white). Find every n such that after a finite number of moves, every unit square on the board has a same color.

Problem 4. (India TST 2016 Day 3 Problem 3) Let n be an odd natural number. We consider an $n \times n$ grid which is made up of n^2 unit squares and $2n(n+1)$ edges. We colour each of these edges either *red* or *blue*. If there are at most n^2 *red* edges, then show that there exists a unit square at least three of whose edges are *blue*.

Problem 5. (EGMO 2016 Day 2 Problem 5) Let k and n be integers such that $k \geq 2$ and $k \leq n \leq 2k-1$. Place rectangular tiles, each of size $1 \times k$, or $k \times 1$ on a $n \times n$ chessboard so that each tile covers exactly k cells and no two tiles overlap. Do this until no further tile can be placed in this way. For each such k and n , determine the minimum number of tiles that such an arrangement may contain.

Problem 6. (IMOSL 2020 C1) Let n be a positive integer. Find the number of permutations $a_1, a_2, \dots, a_n$ of the sequence $1, 2, \dots, n$ satisfying

$$a_1 \leq 2a_2 \leq 3a_3 \leq \dots \leq na_n.$$

Problem 7. (Nordic 2016 P4) King George has decided to connect the 1680 islands in his kingdom by bridges. Unfortunately the rebel movement will destroy two bridges after all the bridges have been built, but not two bridges from the same island. What is the minimal number of bridges the King has to build in order to make sure that it is still possible to travel by bridges between any two of the 1680 islands after the rebel movement has destroyed two bridges?

Problem 8. (BxMO2021 problem 2) Pebbles are placed on the squares of a 2021×2021 board in such a way that each square contains at most one pebble. The pebble set of a square of the board is the collection of all pebbles which are in the same row or column as this square. (A pebble belongs to the pebble set of the square in which it is placed.) What is the least possible number of pebbles on the board if no two squares have the same pebble set?